Practical Fixed-Income Demos

Table of contents

These executable demonstrations connect the lecture models to realistic analyst and portfolio-management workflows. Each case separates observed or supplied inputs, estimated quantities, diagnostics, and the decision that uses the output. The curve and Vasicek cases embed dated official market observations for deterministic offline rendering; other cases clearly identify their market-like teaching inputs. HTML code is folded by default, while results and mathematical conventions remain visible.

Lecture	Workflow	Demo
9	DV01-based Treasury futures hedging and residual risks	Treasury futures hedging workflow
10	Spread duration, DTS, carry, expected loss, and stress tests	Credit-spread risk workflow
11	Treasury futures options and asymmetric portfolio protection	Interest-rate options workflow

The examples are teaching tools rather than trading recommendations. Production valuation requires current market conventions, instrument terms, liquidity inputs, and independent model validation.